advanced_strings

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Strings can do operations on themselves `.lower(),.upper(),.capitalize()`

```
In [1]: "funKY tOwn".capitalize()
```

```
Out[1]: 'Funky town'
```

```
In [2]: "funky tOwn".lower()
```

```
Out[2]: 'funky town'
```

```
In [3]: "fUNKY tOWN".swapcase()
```

```
Out[3]: 'Funky Town'
```

`.split([sep [,maxsplit]])`

```
In [4]: "funKY tOwn".split()
```

```
Out[4]: ['funKY', 'tOwn']
```

```
In [5]: "funKY tOwn".capitalize().split()
```

```
Out[5]: ['Funky', 'town']
```

```
In [6]: [x.capitalize() for x in "funKY tOwn".split()]
```

```
Out[6]: ['Funky', 'Town']
```

```
In [7]: "I want to take you to, funKY tOwn".split("u")
```

```
Out[7]: ['I want to take yo', ' to, f', 'nKY tOwn']
```

```
In [8]: "I want to take you to, funKY tOwn".split("you")
```

```
Out[8]: ['I want to take ', ' to, funKY tOwn']
```

`.strip(),.join(),.replace()`

```
In [9]: csv_string = 'Dog,Cat,Spam,Defenestrate,1, 3.1415   \n\t'
        csv_string.strip()
```

```
Out[9]: 'Dog,Cat,Spam,Defenestrate,1, 3.1415'
```

```
In [10]: clean_list = [x.strip() for x in csv_string.split(",")]
         print(clean_list)
```

```
['Dog', 'Cat', 'Spam', 'Defenestrate', '1', '3.1415']
```

`.join()` allows you to glue a list of strings together with a certain string

```
In [11]: print(",".join(clean_list))
```

```
Dog,Cat,Spam,Defenestrate,1,3.1415
```

```
In [12]: print("\t".join(clean_list))
```

```
Dog	Cat	Spam	Defenestrate	1	3.1415
```

`.replace()` strings in strings

```
In [13]: csv_string = 'Dog,Cat,Spam,Defenestrate,1, 3.1415   \n\t'
         alt_csv = csv_string.strip().replace(' ','')
         print(alt_csv)
```

```
Dog,Cat,Spam,Defenestrate,1,3.1415
```

```
In [14]: print(csv_string.strip().replace(' ','').replace(',','\t'))
```

```
Dog	Cat	Spam	Defenestrate	1	3.1415
```

0.0.1 `.find()`

incredibly useful searching, returning the index of the search

```
In [15]: s = 'My Funny Valentine'
         s.find("y")
```

```
Out[15]: 1
```

```
In [16]: s.find("y",2)
```

```
Out[16]: 7
```

```
In [17]: s[s.find("Funny"):]
```

```
Out[17]: 'Funny Valentine'
```

```
In [18]: s.find("z")

Out[18]: -1

In [19]: ss = [s,"Argentine","American","Quarentine"]
         for thestring in ss:
            if thestring.find("tine") != -1:
               print("'" + str(thestring) + "' contains 'tine'.")

'My Funny Valentine' contains 'tine'.
'Argentine' contains 'tine'.
'Quarentine' contains 'tine'.
```

`string` module exposes useful variables and functions

```
In [20]: import string

In [21]: print(string.ascii_letters)
         print(string.digits)

abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
0123456789
```

0.0.2 String Formatting

the (new) preferred way
is `string.format(value0,value1,....)`

```
In [22]: 'on {0}, I feel {1}'.format("saturday","groovy")

Out[22]: 'on saturday, I feel groovy'

In [23]: 'on {}, I feel {}'.format("saturday","groovy")

Out[23]: 'on saturday, I feel groovy'

In [24]: 'on {0}, I feel {1}'.format(["saturday","groovy"])

         ---------------------------------------------------------------------------

         IndexError                                Traceback (most recent call last)

         <ipython-input-24-37beb7743cdb> in <module>()
         ----> 1 'on {0}, I feel {1}'.format(["saturday","groovy"])


         IndexError: tuple index out of range
```

```
In [25]: 'on {0}, I feel {0}'.format(["saturday","groovy"])

Out[25]: "on ['saturday', 'groovy'], I feel ['saturday', 'groovy']"

In [26]: 'on {0}, I feel {0}'.format("saturday","groovy")

Out[26]: 'on saturday, I feel saturday'
```

you can assign by argument position or by name

```
In [27]: '{desire} to {place}'.format(desire='Fly me',\
                                      place='The Moon')

Out[27]: 'Fly me to The Moon'

In [28]: '{desire} to {place} or else I wont visit {place}.'.format( \
                            desire='Fly me',place='The Moon')

Out[28]: 'Fly me to The Moon or else I wont visit The Moon.'

In [29]: f = {"desire": "I want to take you", "place": "funky town"}

In [30]: '{desire} to {place}'.format(**f)

Out[30]: 'I want to take you to funky town'

In [ ]:
```